



BNU

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Chap 7 Evaluating And Controlling Technology

- o Evaluating Information
- o The Need for Responsible Judgment
- o Computer Models
- o The “Digital Divide”
- o Trends in Computer Access
- o The Global Divide and the Next Billion Users
- o Neo-Luddite Views of Computers, Technology, and Quality of Life
- o Criticisms of Computing Technologies
- o Views of Economics, Nature, and Human Needs Making Decisions About Technology
- o Intelligent Machines and Super intelligent Humans—Or the End of the
- o Human Race?

1. Evaluating Information

- o Ability to assess accuracy, reliability, and relevance of information
- o Important in the age of fake news, misinformation, and AI-generated content

Key evaluation criteria:

- o Source credibility (author, organization)
- o Evidence and references
- o Bias and objectivity
- o Timeliness (updated data)
- o Critical thinking is essential to avoid misleading conclusions

2. The Need for Responsible Judgment

- Technology decisions impact society, privacy, and ethics

Requires:

- Ethical reasoning
- Awareness of social consequences
- Accountability in decision-making

Professionals must balance:

- Innovation vs risk
- Profit vs public welfare

Example: Data privacy vs business analytics

3. Computer Models

Mathematical or simulation-based systems used to ****predict outcomes****

o * Used in:

o

o * Weather forecasting

o * Business decision-making

o * Healthcare and research

o * Advantages:

o

o * Fast processing of complex data

o * Helps in ****decision support****

o * Limitations:

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o * Based on assumptions → may be inaccurate

o * “Garbage in, garbage out” problem

4. The Digital Divide

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o * Gap between people who have access to technology and those who don't

o * Types:

o

o * **Access divide** (devices, internet)

o * **Skill divide** (digital literacy)

o * Causes:

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o * Poverty

o * Lack of infrastructure

o * Education level

o * Effects:

o

o * Inequality in education, jobs, and opportunities

5. Trends in Computer Access

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o * Increasing use of:

o

o * Smartphones

o * Cloud computing

o * Internet services

o * Growth in:

o

o * Remote work

o * Online education

o * Shift from **desktop → mobile-first world**

o * However, inequality still exists in rural vs urban areas

6. The Global Divide and the Next Billion Users

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o * Focus on connecting the ****next billion users**** (mostly in developing countries)

o * Challenges:

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o * Infrastructure

o * Cost of devices and internet

o * Language barriers

o * Opportunities:

o

o * Economic growth

o * Education access

o * Digital inclusion

o * Companies expanding into ****Asia, Africa, and rural regions****

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7. Neo-Luddite Views of Technology

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o * Neo-Luddites criticize rapid technological growth

o * Concerns:

o

o * Loss of jobs due to automation

o * Overdependence on machines

o * Reduced human interaction

o * Believe technology may **harm quality of life**

o * Advocate for **controlled and limited use of technology**

o

8. Criticisms of Computing Technologies

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o * Privacy invasion (data tracking, surveillance)

o * Cybercrime and hacking

o * Job displacement due to automation

o * Health issues:

o

o * Screen addiction

o * Reduced physical activity

o * Environmental concerns:

o

o * E-waste

o * Energy consumption

o

9. Views of Economics, Nature, and Human Needs

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o * Technology affects:

o

o * **Economy** → productivity, jobs, global trade

o * **Nature** → pollution vs green tech

o * **Human needs** → comfort, communication, healthcare

o * Debate:

o

o * Is technology fulfilling real needs or creating artificial ones?

o * Balance required between:

o

o * Growth and sustainability

o

10. Making Decisions About Technology

Requires:

- o * Cost-benefit analysis
- o * Ethical considerations
- o * Risk assessment

Questions to ask:

- o * Who benefits?
- o * Who is harmed?
- o * Is it sustainable?

Involves stakeholders:

- o * Government
- o * Businesses
- o * Society

11. Intelligent Machines and Superintelligent Humans — Or the End of the Human Race?

Concept

- Development of **Artificial Intelligence (AI)** that may surpass human intelligence

Optimistic View

AI can:

- * Improve healthcare
- * Solve global problems
- * Enhance human abilities
- * Concept of **human-AI collaboration**

Pessimistic View

Risks:

- * Loss of human control
- * Job elimination
- * Ethical concerns
- * Fear of **superintelligent AI dominating humans**

11. Intelligent Machines and Superintelligent Humans — Or the End of the Human Race?

Key Debate

Will AI:

- o * **Enhance humanity?**
- o * Or **replace it?**

Conclusion

Future depends on:

- o * Responsible development
- o * Ethical guidelines
- o * Human control over technology